

(54) DETECTION OF PARALLEL OPERATING WI-FI NETWORKS	Publication Classification
(71) Applicant: PLUME DESIGN, INC. , Palo Alto, CA (US)	(51) Int. Cl.
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	(52) U.S. Cl.
	CPC <i>H04W 48/16</i> (2013.01); <i>H04W 16/20</i> (2013.01); <i>H04W 68/005</i> (2013.01); <i>H04W 84/12</i> (2013.01)
	(57) ABSTRACT

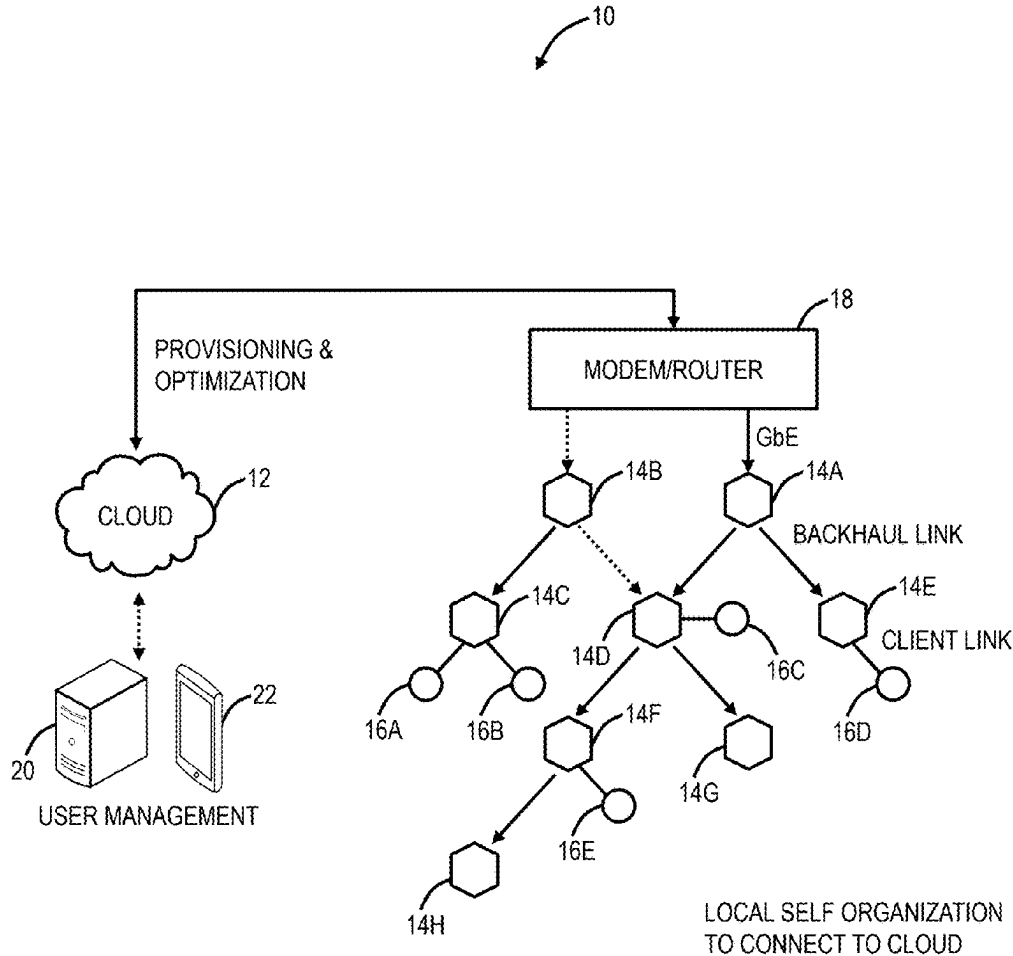
(21) Appl. No.: **19/212,855**

(22) Filed: **May 20, 2025**

Related U.S. Application Data

(63) Continuation of application No. 18/489,899, filed on Oct. 19, 2023, now Pat. No. 12,335,852, which is a continuation of application No. 17/745,404, filed on May 16, 2022, now Pat. No. 11,832,170, which is a continuation of application No. 17/078,196, filed on Oct. 23, 2020, now Pat. No. 11,368,903.

Systems and methods of detecting a parallel Wi-Fi network or a parallel Wi-Fi access point include operating a new Wi-Fi network at a location; analyzing Wi-Fi at the location; determining whether there is a parallel Wi-Fi network or a parallel Wi-Fi access point operating at the location with the new Wi-Fi network based on the analyzing; and, responsive to determining there is the parallel Wi-Fi network or the parallel Wi-Fi access point at the location, performing one or more of i) causing resolution of the parallel Wi-Fi network or the parallel Wi-Fi access point and ii) providing a notification to a user associated with the location.



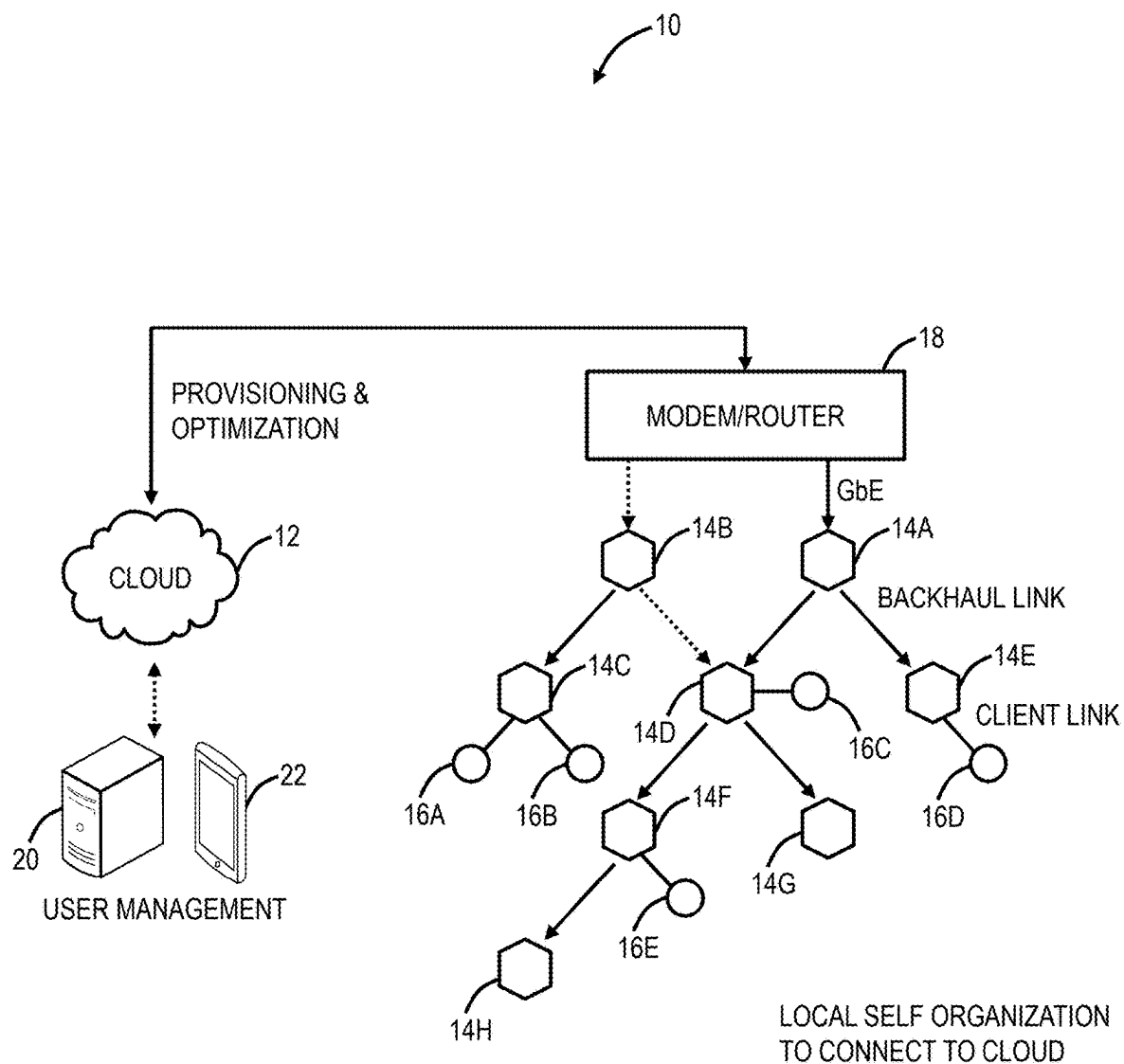


FIG. 1

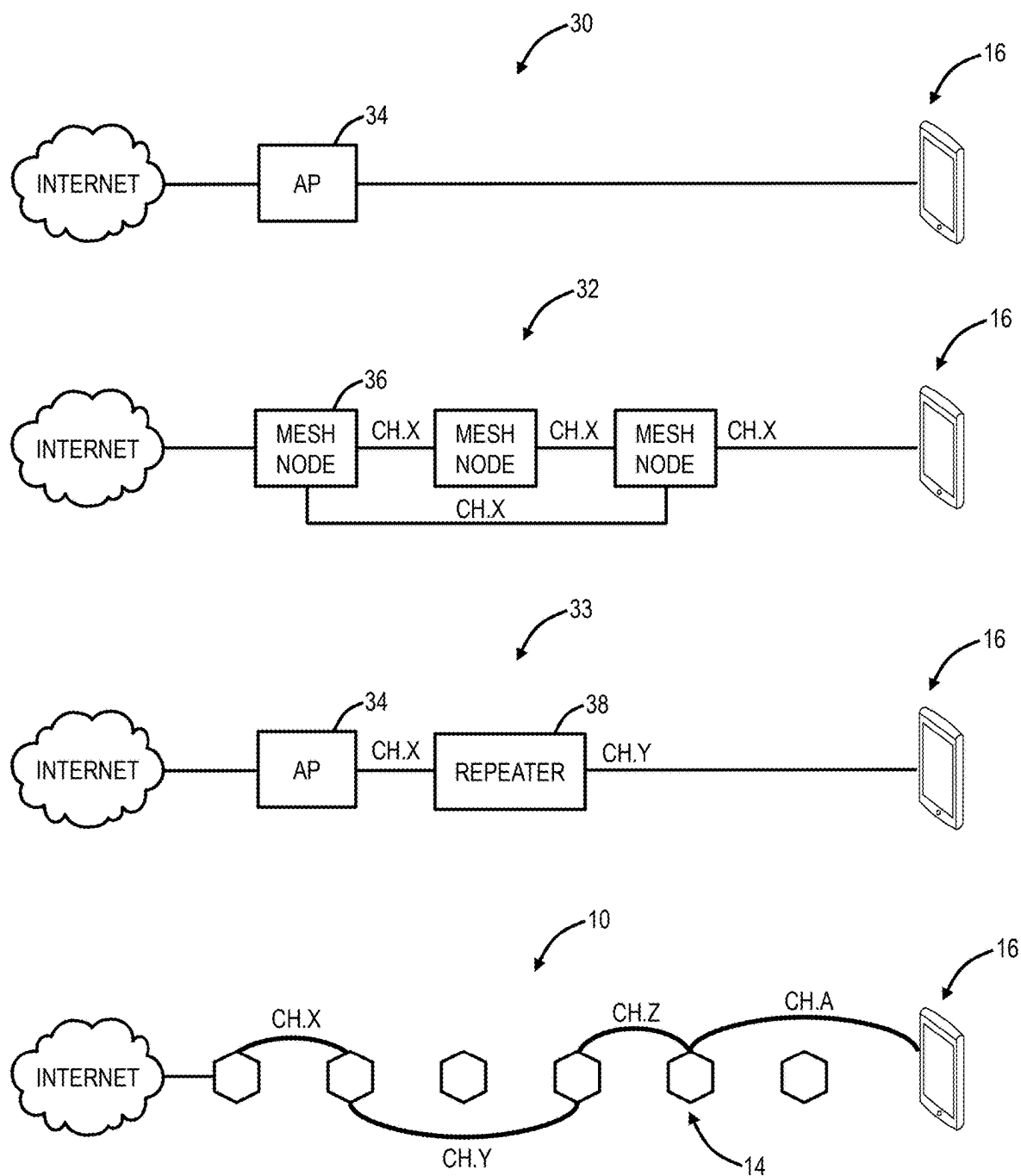


FIG. 2

FIG. 3

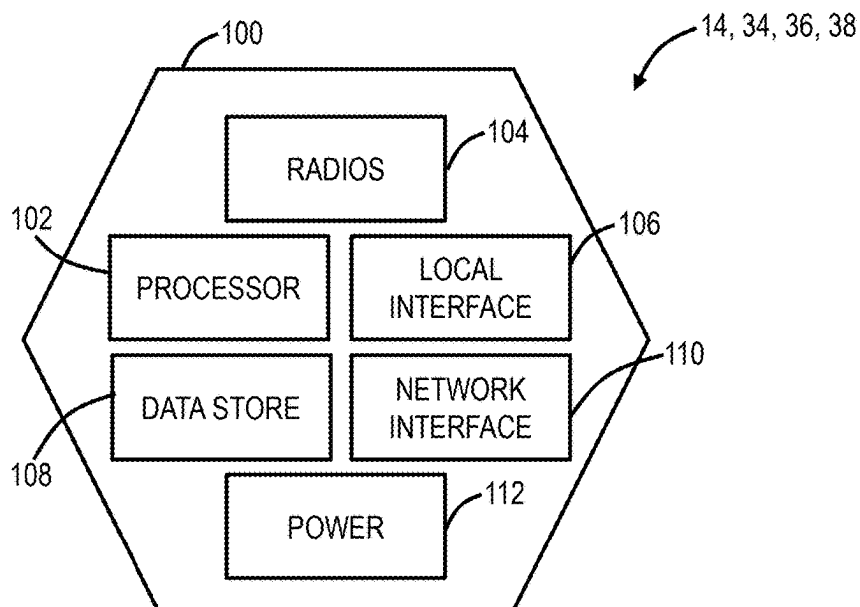
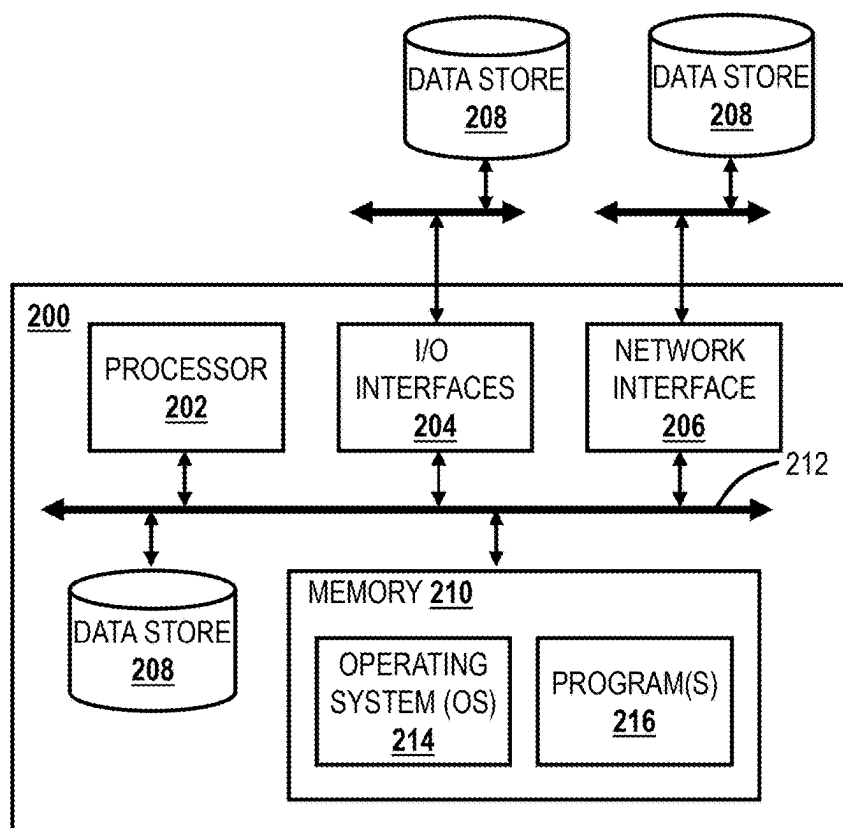
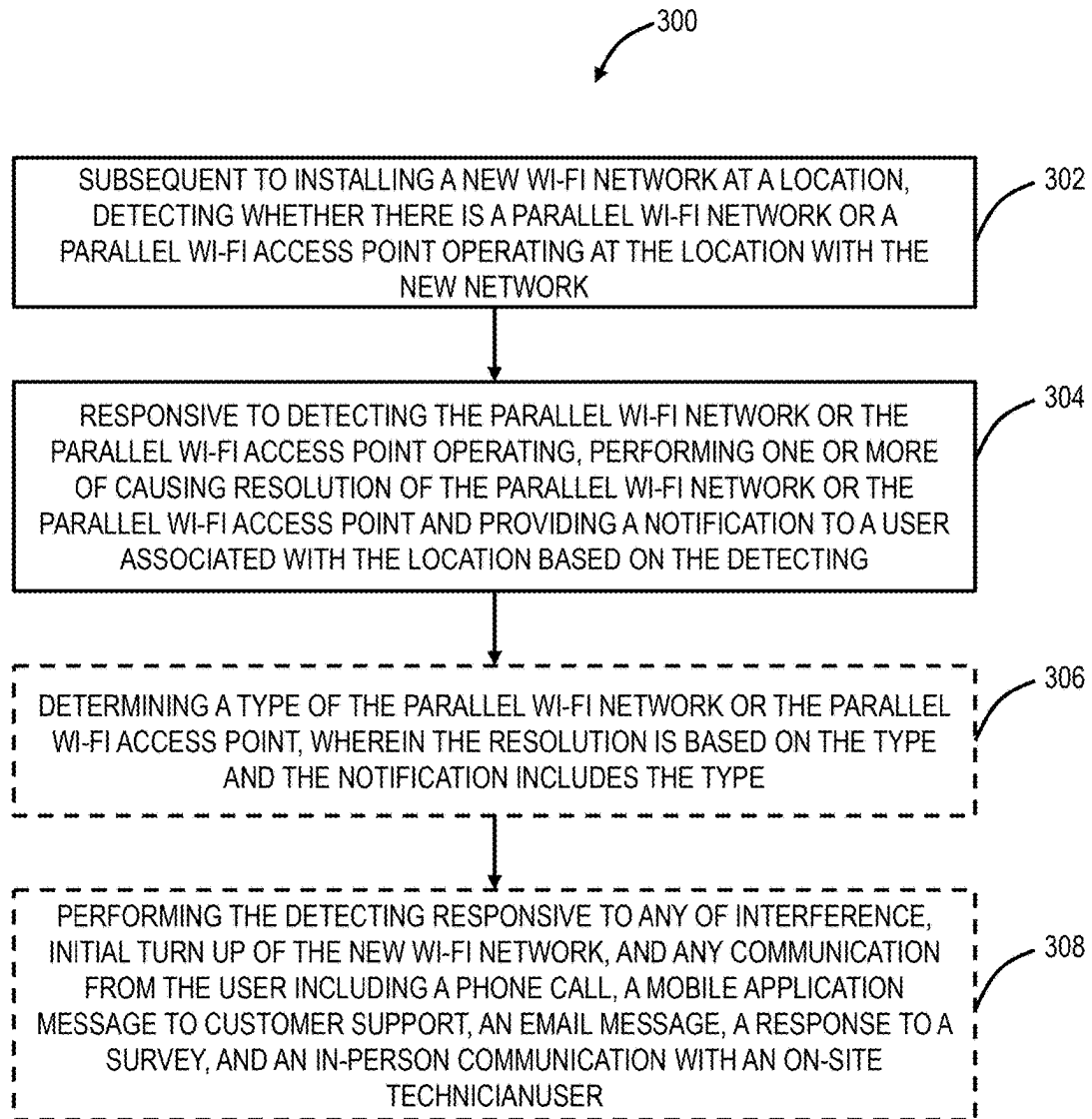


FIG. 4



**FIG. 5**

DETECTION OF PARALLEL OPERATING WI-FI NETWORKS

CROSS-REFERENCE TO RELATED APPLICATION(S)

[0001] The present disclosure is a continuation of U.S. patent application Ser. No. 18/489,899, filed Oct. 19, 2023, which is a continuation of U.S. patent application Ser. No. 17/745,404, filed May 16, 2022 (now U.S. Pat. No. 11,832,170), which is a continuation of U.S. patent application Ser. No. 17/078,196, filed Oct. 23, 2020, now U.S. Pat. No. 11,368,903, the contents of which are incorporated by reference in their entirety.

FIELD OF THE DISCLOSURE

[0002] The present disclosure generally relates to networking systems and methods. More particularly, the present disclosure relates to detection of parallel operating Wi-Fi networks.

BACKGROUND OF THE DISCLOSURE

[0003] Conventional Wi-Fi networks are deployed by users in their residences to provide network connectivity to various devices (e.g., mobile devices, tablets, televisions, Internet of Things (IoT) devices, laptops, media players, and the like). The users obtain network connectivity from their service provider, e.g., Multiple Systems Operators (MSOs), wireless providers, telecom providers, etc. Often the users have an existing Wi-Fi network and add a new, upgraded Wi-Fi network to improve functionality. The existing Wi-Fi network can include Wi-Fi that is integrated into a modem/router/gateway from the service provider. When the new, upgraded Wi-Fi network is added, it is common to use the same Service Set Identifier (SSID) so that the user does not have to change the Wi-Fi configuration of various Wi-Fi client devices. For example, a typical home can have tens of devices that connect, and changing the SSID for each of these is far more complicated than keeping the same SSID when upgrading Wi-Fi equipment. Problematically, users often do not decommission or turn off the existing Wi-Fi network. This is especially common where the existing Wi-Fi network is integrated into the modem/router/gateway from the service provider. This leads to a parallel Wi-Fi network with the same SSID operating in the same location.

BRIEF SUMMARY OF THE DISCLOSURE

[0004] The present disclosure relates to detection of parallel operating Wi-Fi networks. Specifically, the objective is to identify parallel operating Wi-Fi networks to remove one of the networks to improve performance. The present disclosure includes cloud-based Wi-Fi network management that can include various processes to detect parallel Wi-Fi networks. Once detected, a customer can be notified with instructions on how to remove the parallel Wi-Fi network, leading to improved performance on the remaining Wi-Fi network. Examples of parallel networks include a so-called “evil twin” which is another Wi-Fi network, operating in parallel, with the same SSID; a so-called “evil brother” which is another Wi-Fi network, operating in parallel, with a different SSID; and a so-called “lonely network” which is a new Wi-Fi network with zero or very few clients, presumably because many other devices are on a parallel network.

[0005] In various embodiments, the present disclosure includes a method including steps, a controller configured to implement the steps, and a non-transitory computer-readable medium with instructions that cause a processor to perform the steps. The steps relate to detecting a parallel Wi-Fi network or a parallel Wi-Fi access point. The steps include, subsequent to installing a new Wi-Fi network at a location, detecting whether there is a parallel Wi-Fi network or a parallel Wi-Fi access point operating at the location with the new network; and, responsive to detecting the parallel Wi-Fi network or the parallel Wi-Fi access point operating, performing one or more of causing resolution of the parallel Wi-Fi network or the parallel Wi-Fi access point and providing a notification to a user associated with the location based on the detecting.

[0006] The steps can further include determining a type of the parallel Wi-Fi network or the parallel Wi-Fi access point, wherein the resolution is based on the type and the notification includes the type. The steps can further include performing the detecting responsive to any of interference, initial turn up of the new Wi-Fi network, and any communication from the user including a phone call, a mobile application message to customer support, an email message, a response to a survey, and an in-person communication with an on-site technician. The steps can further include performing the detecting responsive to detecting the new Wi-Fi network has zero or very few Wi-Fi client devices connected thereto, presumably because the Wi-Fi client devices are on the parallel Wi-Fi network or the parallel Wi-Fi access point.

[0007] The notification can include details for the user to turn off the parallel Wi-Fi network or the parallel Wi-Fi access point. The notification can include details for the user to remove profiles for the parallel Wi-Fi network or the parallel Wi-Fi access point in Wi-Fi client devices at the location. The resolution can include disabling the parallel Wi-Fi network or the parallel Wi-Fi access point remotely. The resolution can include notifying a service provider of the user to cause disabling the parallel Wi-Fi network or the parallel Wi-Fi access point. Subsequent to the performing, repeating the detecting to determine if the parallel Wi-Fi network or the parallel Wi-Fi access point remain.

[0008] The steps can further include providing a dashboard that displays locations where the parallel Wi-Fi network or the parallel Wi-Fi access point have been detected. The steps can further include causing initiation of support for a service provider based on the detecting.

[0009] The parallel Wi-Fi network or the parallel Wi-Fi access point can have a same Service Set Identifier (SSID) or a different SSID as the new network. The detecting can be based on an access point in the new Wi-Fi network detecting another access point that is not part of the new Wi-Fi network but that has the same SSID. The detecting can be based on detecting Wi-Fi client devices disappearing and reappearing on the new Wi-Fi network, factoring whether those Wi-Fi client devices are mobile or non-mobile devices. The detecting can be based on a combination of an access point in the new Wi-Fi network detecting another access point that is not part of the new Wi-Fi network but that has the same SSID, and detecting Wi-Fi client devices disappearing and reappearing on the new Wi-Fi network, with at least some of the Wi-Fi client devices being non-mobile devices.

BRIEF DESCRIPTION OF THE DRAWINGS

[0010] The present disclosure is illustrated and described herein with reference to the various drawings, in which like reference numbers are used to denote like system components/method steps, as appropriate, and in which:

[0011] FIG. 1 is a network diagram of a distributed Wi-Fi system with cloud-based control and management;

[0012] FIG. 2 is a network diagram of differences in the operation of the distributed Wi-Fi system of FIG. 1 relative to a conventional single access point system, a Wi-Fi mesh network, and a Wi-Fi repeater network;

[0013] FIG. 3 is a block diagram of functional components of the access point;

[0014] FIG. 4 is a block diagram of functional components of the server, the Wi-Fi client device, or the user device; and

[0015] FIG. 5 is a flowchart of a parallel network detection and resolution process.

DETAILED DESCRIPTION OF THE DISCLOSURE

[0016] The present disclosure relates to detection of parallel operating Wi-Fi networks. Specifically, the objective is to identify parallel operating Wi-Fi networks to remove one of the networks to improve performance. The present disclosure includes cloud-based Wi-Fi network management that can include various processes to detect parallel Wi-Fi networks. Once detected, a customer can be notified with instructions on how to remove the parallel Wi-Fi network, leading to improved performance on the remaining Wi-Fi network. Examples of parallel networks include a so-called “evil twin” which is another Wi-Fi network, operating in parallel, with the same SSID; a so-called “evil brother” which is another Wi-Fi network, operating in parallel, with a different SSID; and a so-called “lonely network” which is a new Wi-Fi network with zero or very few clients, presumably because many other devices are on a parallel network.

Distributed Wi-Fi System

[0017] FIG. 1 is a network diagram of a distributed Wi-Fi system 10 with control via a cloud 12 service. The distributed Wi-Fi system 10 can operate in accordance with the IEEE 802.11 protocols and variations thereof. The distributed Wi-Fi system 10 includes a plurality of access points 14 (labeled as access points 14A-14H), which can be distributed throughout a location, such as a residence, office, or the like. That is, the distributed Wi-Fi system 10 contemplates operation in any physical location where it is inefficient or impractical to service with a single access point, repeaters, or a mesh system. As described herein, the distributed Wi-Fi system 10 can be referred to as a network, a system, a Wi-Fi network, a Wi-Fi system, a cloud-based system, etc. The access points 14 can be referred to as nodes, access points, Wi-Fi nodes, Wi-Fi access points, etc. The objective of the access points 14 is to provide network connectivity to Wi-Fi client devices 16 (labeled as Wi-Fi client devices 16A-16E). The Wi-Fi client devices 16 can be referred to as client devices, user devices, clients, Wi-Fi clients, Wi-Fi devices, etc.

[0018] In a typical residential deployment, the distributed Wi-Fi system 10 can include between 3 to 12 access points or more in a home. A large number of access points 14 (which can also be referred to as nodes in the distributed Wi-Fi system 10) ensures that the distance between any

access point 14 is always small, as is the distance to any Wi-Fi client device 16 needing Wi-Fi service. That is, an objective of the distributed Wi-Fi system 10 can be for distances between the access points 14 to be of similar size as distances between the Wi-Fi client devices 16 and the associated access point 14. Such small distances ensure that every corner of a consumer's home is well covered by Wi-Fi signals. It also ensures that any given hop in the distributed Wi-Fi system 10 is short and goes through few walls. This results in very strong signal strengths for each hop in the distributed Wi-Fi system 10, allowing the use of high data rates, and providing robust operation. Note, those skilled in the art will recognize the Wi-Fi client devices 16 can be mobile devices, tablets, computers, consumer electronics, home entertainment devices, televisions, IoT devices, or any network-enabled device. For external network connectivity, one or more of the access points 14 can be connected to a modem/router 18, which can be a cable modem, Digital Subscriber Loop (DSL) modem, or any device providing external network connectivity to the physical location associated with the distributed Wi-Fi system 10.

[0019] While providing excellent coverage, a large number of access points 14 (nodes) presents a coordination problem. Getting all the access points 14 configured correctly and communicating efficiently requires centralized control. This cloud 12 service can provide control via servers 20 that can be reached across the Internet and accessed remotely, such as through an application (“app”) running on a user device 22. The running of the distributed Wi-Fi system 10, therefore, becomes what is commonly known as a “cloud service.” The servers 20 are configured to receive measurement data, to analyze the measurement data, and to configure the access points 14 in the distributed Wi-Fi system 10 based thereon, through the cloud 12. The servers 20 can also be configured to determine which access point 14 each of the Wi-Fi client devices 16 connect (associate) with. That is, in an example aspect, the distributed Wi-Fi system 10 includes cloud-based control (with a cloud-based controller or cloud service in the cloud) to optimize, configure, and monitor the operation of the access points 14 and the Wi-Fi client devices 16. This cloud-based control is contrasted with a conventional operation that relies on a local configuration, such as by logging in locally to an access point. In the distributed Wi-Fi system 10, the control and optimization does not require local login to the access point 14, but rather the user device 22 (or a local Wi-Fi client device 16) communicating with the servers 20 in the cloud 12, such as via a disparate network (a different network than the distributed Wi-Fi system 10) (e.g., LTE, another Wi-Fi network, etc.).

[0020] The access points 14 can include both wireless links and wired links for connectivity. In the example of FIG. 1, the access point 14A has an example gigabit Ethernet (GbE) wired connection to the modem/router 18. Optionally, the access point 14B also has a wired connection to the modem/router 18, such as for redundancy or load balancing. Also, the access points 14A, 14B can have a wireless connection to the modem/router 18. The access points 14 can have wireless links for client connectivity (referred to as a client link) and for backhaul (referred to as a backhaul link). The distributed Wi-Fi system 10 differs from a conventional Wi-Fi mesh network in that the client links and the backhaul links do not necessarily share the same Wi-Fi channel, thereby reducing interference. That is, the access

points **14** can support at least two Wi-Fi wireless channels—which can be used flexibly to serve either the client link or the backhaul link and may have at least one wired port for connectivity to the modem/router **18**, or for connection to other devices. In the distributed Wi-Fi system **10**, only a small subset of the access points **14** require direct connectivity to the modem/router **18** with the non-connected access points **14** communicating with the modem/router **18** through the backhaul links back to the connected access points **14**.

Distributed Wi-Fi System Compared to Conventional Wi-Fi Systems

[0021] FIG. 2 is a network diagram of differences in the operation of the distributed Wi-Fi system **10** relative to a conventional single access point system **30**, a Wi-Fi mesh network **32**, and a Wi-Fi repeater network **33**. The single access point system **30** relies on a single, high-powered access point **34**, which may be centrally located to serve all Wi-Fi client devices **16** in a location (e.g., house). Again, as described herein, in a typical residence, the single access point system **30** can have several walls, floors, etc. between the access point **34** and the Wi-Fi client devices **16**. Plus, the single access point system **30** operates on a single channel, leading to potential interference from neighboring systems. The Wi-Fi mesh network **32** solves some of the issues with the single access point system **30** by having multiple mesh nodes **36**, which distribute the Wi-Fi coverage. Specifically, the Wi-Fi mesh network **32** operates based on the mesh nodes **36** being fully interconnected with one another, sharing a channel such as a channel X between each of the mesh nodes **36** and the Wi-Fi client device **16**. That is, the Wi-Fi mesh network **32** is a fully interconnected grid, sharing the same channel, and allowing multiple different paths between the mesh nodes **36** and the Wi-Fi client device **16**. However, since the Wi-Fi mesh network **32** uses the same backhaul channel, every hop between source points divides the network capacity by the number of hops taken to deliver the data. For example, if it takes three hops to stream a video to a Wi-Fi client device **16**, the Wi-Fi mesh network **32** is left with only $\frac{1}{3}$ the capacity. The Wi-Fi repeater network **33** includes the access point **34** coupled wirelessly to a Wi-Fi repeater **38**. The Wi-Fi repeater network **33** is a star topology where there is at most one Wi-Fi repeater **38** between the access point **14** and the Wi-Fi client device **16**. From a channel perspective, the access point **34** can communicate to the Wi-Fi repeater **38** on a first channel, Ch. X, and the Wi-Fi repeater **38** can communicate to the Wi-Fi client device **16** on a second channel, Ch. Y.

[0022] The distributed Wi-Fi system **10** solves the problem with the Wi-Fi mesh network **32** of requiring the same channel for all connections by using a different channel or band for the various hops (note, some hops may use the same channel/band, but it is not required), to prevent slowing down the Wi-Fi speed. For example, the distributed Wi-Fi system **10** can use different channels/bands between access points **14** and between the Wi-Fi client device **16** (e.g., Chs. X, Y, Z, A), and also, the distributed Wi-Fi system **10** does not necessarily use every access point **14**, based on configuration and optimization by the cloud **12**. The distributed Wi-Fi system **10** solves the problems of the single access point system **30** by providing multiple access points **14**. The distributed Wi-Fi system **10** is not constrained to a star topology as in the Wi-Fi repeater network **33**, which at most allows two wireless hops between the Wi-Fi client device **16**

and a gateway. Also, the distributed Wi-Fi system **10** forms a tree topology where there is one path between the Wi-Fi client device **16** and the gateway, but which allows for multiple wireless hops, unlike the Wi-Fi repeater network **33**.

[0023] Wi-Fi is a shared, simplex protocol meaning only one conversation between two devices can occur in the network at any given time, and if one device is talking the others need to be listening. By using different Wi-Fi channels, multiple simultaneous conversations can happen simultaneously in the distributed Wi-Fi system **10**. By selecting different Wi-Fi channels between the access points **14**, interference and congestion are avoided. The server **20** through the cloud **12** automatically configures the access points **14** in an optimized channel hop solution. The distributed Wi-Fi system **10** can choose routes and channels to support the ever-changing needs of consumers and their Wi-Fi client devices **16**. The distributed Wi-Fi system **10** approach is to ensure Wi-Fi signals do not need to travel far—either for backhaul or client connectivity. Accordingly, the Wi-Fi signals remain strong and avoid interference by communicating on the same channel as in the Wi-Fi mesh network **32** or with Wi-Fi repeaters. In an example aspect, the servers **20** in the cloud **12** are configured to optimize channel selection for the best user experience.

[0024] Of note, the present disclosure for intelligent monitoring is not limited to the distributed Wi-Fi system **10** but contemplates any of the Wi-Fi networks **10**, **30**, **32**, **33**, with monitoring through the cloud **12**. For example, different vendors can make access points **14**, **34**, mesh nodes **36**, repeaters **38**, etc. However, it is possible for unified control via the cloud using standardized techniques for communication with the cloud **12**. One such example includes OpenSync, sponsored by the Applicant of the present disclosure and described at www.opensync.io/documentation. OpenSync is cloud-agnostic open-source software for the delivery, curation, and management of services for the modern home. That is, this provides standardization of the communication between devices and the cloud **12**. OpenSync acts as silicon, Customer Premises Equipment (CPE), and cloud-agnostic connection between the in-home hardware devices and the cloud **12**. This is used to collect measurements and statistics from the connected Wi-Fi client devices **16** and network management elements, and to enable customized connectivity services.

Cloud-Based Wi-Fi Management

[0025] Conventional Wi-Fi systems utilize local management, such as where a user on the Wi-Fi network connects to a designated address (e.g., 192.168.1.1, etc.). The distributed Wi-Fi system **10** is configured for cloud-based management via the servers **20** in the cloud **12**. Also, the single access point system **30**, the Wi-Fi mesh network **32**, and the Wi-Fi repeater network **33** can support cloud-based management as described above. For example, the APs **34** and/or the mesh nodes **36** can be configured to communicate with the servers **20** in the cloud **12**. This configuration can be through a software agent installed in each device or the like, e.g., OpenSync. As described herein, cloud-based management includes reporting of Wi-Fi related performance metrics to the cloud **12** as well as receiving Wi-Fi-related configuration parameters from the cloud **12**. The systems and methods contemplate use with any Wi-Fi system (i.e., the distributed Wi-Fi system **10**, the single access point

system **30**, the Wi-Fi mesh network **32**, and the Wi-Fi repeater network **33**, etc.), including systems that only support reporting of Wi-Fi related performance metrics (and not supporting cloud-based configuration).

[0026] The cloud **12** utilizes cloud computing systems and methods abstract away physical servers, storage, networking, etc. and instead offer these as on-demand and elastic resources. The National Institute of Standards and Technology (NIST) provides a concise and specific definition which states cloud computing is a model for enabling convenient, on-demand network access to a shared pool of configurable computing resources (e.g., networks, servers, storage, applications, and services) that can be rapidly provisioned and released with minimal management effort or service provider interaction. Cloud computing differs from the classic client-server model by providing applications from a server that are executed and managed by a client's web browser or the like, with no installed client version of an application required. Centralization gives cloud service providers complete control over the versions of the browser-based and other applications provided to clients, which removes the need for version upgrades or license management on individual client computing devices. The phrase SaaS is sometimes used to describe application programs offered through cloud computing. A common shorthand for a provided cloud computing service (or even an aggregation of all existing cloud services) is "the cloud."

Access Point

[0027] FIG. 3 is a block diagram of functional components of the access point **14**, **34**, **36**, **38**. The access point **14** includes a physical form factor **100** which contains a processor **102**, a plurality of radios **104**, a local interface **106**, a data store **108**, a network interface **110**, and power **112**. It should be appreciated by those of ordinary skill in the art that FIG. 3 depicts the access point **14** in an oversimplified manner, and a practical embodiment may include additional components and suitably configured processing logic to support features described herein or known or conventional operating features that are not described in detail herein.

[0028] In an exemplary embodiment, the form factor **100** is a compact physical implementation where the access point **14** directly plugs into an electrical socket and is physically supported by the electrical plug connected to the electrical socket. This compact physical implementation is ideal for a large number of access points **14** distributed throughout a residence. The processor **102** is a hardware device for executing software instructions. The processor **102** can be any custom made or commercially available processor, a central processing unit (CPU), an auxiliary processor among several processors associated with the mobile device **300**, a semiconductor-based microprocessor (in the form of a microchip or chip set), or generally any device for executing software instructions. When the access point **14** is in operation, the processor **102** is configured to execute software stored within memory or the data store **108**, to communicate data to and from the memory or the data store **108**, and to generally control operations of the access point **14** pursuant to the software instructions. In an exemplary embodiment, the processor **102** may include a mobile-optimized processor such as optimized for power consumption and mobile applications.

[0029] The radios **104** enable wireless communication in the distributed Wi-Fi system **10**. The radios **104** can operate

according to the IEEE 802.11 standard. The radios **104** include address, control, and/or data connections to enable appropriate communications on the distributed Wi-Fi system **10**. As described herein, the access point **14** includes a plurality of radios to support different links, i.e., backhaul links and client links. The optimization **70** determines the configuration of the radios **104** such as bandwidth, channels, topology, etc. In an exemplary embodiment, the access points **14** support dual-band operation simultaneously operating 2.4 GHz and 5 GHz 2x2 MIMO 802.11b/g/n/ac radios having operating bandwidths of 20/40 MHz for 2.4 GHz and 20/40/80 MHz for 5 GHz. For example, the access points **14** can support IEEE 802.11AC1200 gigabit Wi-Fi (300+ 867Mbps).

[0030] The local interface **106** is configured for local communication to the access point **14** and can be either a wired connection or wireless connection such as Bluetooth or the like. Since the access points **14** are configured via the cloud **12**, an onboarding process is required to first establish connectivity for a newly turned on access point **14**. In an exemplary embodiment, the access points **14** can also include the local interface **106** allowing connectivity to the user device **22** (or a Wi-Fi client device **16**) for onboarding to the distributed Wi-Fi system **10** such as through an app on the user device **22**. The data store **108** is used to store data. The data store **108** may include any of volatile memory elements (e.g., random access memory (RAM, such as DRAM, SRAM, SDRAM, and the like)), nonvolatile memory elements (e.g., ROM, hard drive, tape, CDROM, and the like), and combinations thereof. Moreover, the data store **108** may incorporate electronic, magnetic, optical, and/or other types of storage media.

[0031] The network interface **110** provides wired connectivity to the access point **14**. The network interface **104** may be used to enable the access point **14** communicate to the modem/router **18**. Also, the network interface **104** can be used to provide local connectivity to a Wi-Fi client device **16** or user device **22**. For example, wiring in a device to an access point **14** can provide network access to a device which does not support Wi-Fi. In an exemplary embodiment, all of the access points **14** in the distributed Wi-Fi system **10** include the network interface **110**. In another exemplary embodiment, select access points **14** which connect to the modem/router **18** or require local wired connections have the network interface **110**. The network interface **110** may include, for example, an Ethernet card or adapter (e.g., 10BaseT, Fast Ethernet, Gigabit Ethernet, 10GbE). The network interface **110** may include address, control, and/or data connections to enable appropriate communications on the network.

[0032] The processor **102** and the data store **108** can include software and/or firmware which essentially controls the operation of the access point **14**, data gathering and measurement control, data management, memory management, and communication and control interfaces with the server **20** via the cloud. The processor **102** and the data store **108** may be configured to implement the various processes, algorithms, methods, techniques, etc. described herein.

Cloud Server and User Device

[0033] FIG. 4 is a block diagram of functional components of the server **20**, the Wi-Fi client device **16**, or the user device **22**. FIG. 4 illustrates functional components which can form any of the Wi-Fi client device **16**, the server **20**, the

user device **22**, or any general processing device. The server **20** may be a digital computer that, in terms of hardware architecture, generally includes a processor **202**, input/output (I/O) interfaces **204**, a network interface **206**, a data store **208**, and memory **210**. It should be appreciated by those of ordinary skill in the art that FIG. 4 depicts the server **20** in an oversimplified manner, and a practical embodiment may include additional components and suitably configured processing logic to support features described herein or known or conventional operating features that are not described in detail herein.

[0034] The components (**202**, **204**, **206**, **208**, and **210**) are communicatively coupled via a local interface **212**. The local interface **212** may be, for example, but not limited to, one or more buses or other wired or wireless connections, as is known in the art. The local interface **212** may have additional elements, which are omitted for simplicity, such as controllers, buffers (caches), drivers, repeaters, and receivers, among many others, to enable communications. Further, the local interface **212** may include address, control, and/or data connections to enable appropriate communications among the aforementioned components.

[0035] The processor **202** is a hardware device for executing software instructions. The processor **202** may be any custom made or commercially available processor, a central processing unit (CPU), an auxiliary processor among several processors associated with the server **20**, a semiconductor-based microprocessor (in the form of a microchip or chip set), or generally any device for executing software instructions. When the server **20** is in operation, the processor **202** is configured to execute software stored within the memory **210**, to communicate data to and from the memory **210**, and to generally control operations of the server **20** pursuant to the software instructions. The I/O interfaces **204** may be used to receive user input from and/or for providing system output to one or more devices or components. User input may be provided via, for example, a keyboard, touchpad, and/or a mouse. System output may be provided via a display device and a printer (not shown). I/O interfaces **204** may include, for example, a serial port, a parallel port, a small computer system interface (SCSI), a serial ATA (SATA), a fibre channel, Infiniband, iSCSI, a PCI Express interface (PCI-x), an infrared (IR) interface, a radio frequency (RF) interface, and/or a universal serial bus (USB) interface.

[0036] The network interface **206** may be used to enable the server **20** to communicate on a network, such as the cloud **12**. The network interface **206** may include, for example, an Ethernet card or adapter (e.g., 10BaseT, Fast Ethernet, Gigabit Ethernet, 10GbE) or a wireless local area network (WLAN) card or adapter (e.g., 802.11a/b/g/n/ac). The network interface **206** may include address, control, and/or data connections to enable appropriate communications on the network. A data store **208** may be used to store data. The data store **208** may include any of volatile memory elements (e.g., random access memory (RAM, such as DRAM, SRAM, SDRAM, and the like)), nonvolatile memory elements (e.g., ROM, hard drive, tape, CDROM, and the like), and combinations thereof. Moreover, the data store **208** may incorporate electronic, magnetic, optical, and/or other types of storage media. In one example, the data store **208** may be located internal to the server **20** such as, for example, an internal hard drive connected to the local interface **212** in the server **20**. Additionally, in another

embodiment, the data store **208** may be located external to the server **20** such as, for example, an external hard drive connected to the I/O interfaces **204** (e.g., SCSI or USB connection). In a further embodiment, the data store **208** may be connected to the server **20** through a network, such as, for example, a network attached file server.

[0037] The memory **210** may include any of volatile memory elements (e.g., random access memory (RAM, such as DRAM, SRAM, SDRAM, etc.)), nonvolatile memory elements (e.g., ROM, hard drive, tape, CDROM, etc.), and combinations thereof. Moreover, the memory **210** may incorporate electronic, magnetic, optical, and/or other types of storage media. Note that the memory **210** may have a distributed architecture, where various components are situated remotely from one another but can be accessed by the processor **202**. The software in memory **210** may include one or more software programs, each of which includes an ordered listing of executable instructions for implementing logical functions. The software in the memory **210** includes a suitable operating system (O/S) **214** and one or more programs **216**. The operating system **214** essentially controls the execution of other computer programs, such as the one or more programs **216**, and provides scheduling, input-output control, file and data management, memory management, and communication control and related services. The one or more programs **216** may be configured to implement the various processes, algorithms, methods, techniques, etc. described herein.

Definitions

[0038] Again, the present disclosure relates to detection of parallel operating Wi-Fi networks. Specifically, the objective is to identify parallel operating Wi-Fi networks to remove one of the networks or access points to improve performance. The following terminology is utilized herein.

[0039] Evil twin is another Wi-Fi network, operating in parallel at the same location, with the same SSID. Here, the location can be a residence. There is a Wi-Fi network of interest installed (a “new” Wi-Fi network) and there is another Wi-Fi network that is operating in parallel (a “parallel” Wi-Fi network or the evil twin). Each of these Wi-Fi networks can be any of the Wi-Fi networks **10**, **30**, **32**, **33**. For example, the evil twin can be the Wi-Fi network **30**, and the new Wi-Fi network can be the Wi-Fi network **10**, **32**, **33**, etc. Other embodiments are contemplated. The typical case here is the parallel Wi-Fi network is operating, such as integrated with the modem/router **18** or another configuration. The parallel Wi-Fi network is typically installed by a service provider (e.g., cable company). The new Wi-Fi network is typically installed later, i.e., an upgrade. Again, a typical use case is the owner installs the new Wi-Fi network to upgrade or provide better coverage. As such, this is typically a multi-access point configuration, such as any of the Wi-Fi networks **10**, **32**, **33**. For convenience, users typically add the new Wi-Fi network with an existing SSID as this reduces configuration of the Wi-Fi client devices **16**. Disadvantageously, often users leave the parallel Wi-Fi network as-is leading to performance issues, namely two parallel Wi-Fi networks operating at the same location (residence).

[0040] An evil brother is an access point **14**, **34**, **36**, **38** that is operating at a same location as a new Wi-Fi network, but with a different SSID. The evil brother can be a single access point or an entirely separate Wi-Fi network. That is, the evil

brother is the same as the evil twin except that it has a separate SSID. A use case here can be where the new Wi-Fi network is added after the fact with a different SSID from the parallel network.

[0041] Thus, an evil twin is a parallel Wi-Fi network or a parallel access point operating at a same location as a new Wi-Fi network with the same SSID. An evil brother is a parallel Wi-Fi network or parallel access point operating at a same location as the new Wi-Fi network with a different SSID. The new Wi-Fi network is referred to as “new” because it has been added after the parallel network, presumably for an upgrade.

[0042] Finally, a lonely network is a result of having the parallel Wi-Fi network or parallel access point at the same location. That is a lonely is a new Wi-Fi network with zero or very few Wi-Fi client devices **16**, presumably because many other devices are on a parallel network.

[0043] Of course, there is no need to operate parallel Wi-Fi networks at the same location. The objective of the present disclosure is to provide techniques for identifying the parallel networks for the purpose of remediation, specifically removing, turning off, and/or decommissioning the parallel networks.

Parallel Network Detection and Resolution

[0044] FIG. 5 is a flowchart of a parallel network detection and resolution process **300**. The parallel network detection process **300** contemplates implementation as a method with steps, via a server **200** or the like with a processor configured to implement the steps, and via a non-transitory computer-readable medium having computer readable code stored thereon for programming a processor to perform the steps.

[0045] The steps include, subsequent to installing a new Wi-Fi network at a location, detecting whether there is a parallel Wi-Fi network or a parallel Wi-Fi access point operating at the location with the new network (step **302**); and responsive to detecting the parallel Wi-Fi network or the parallel Wi-Fi access point operating, performing one or more of causing resolution of the parallel Wi-Fi network or the parallel Wi-Fi access point and providing a notification to a user associated with the location based on the detecting (step **304**). The steps can further include determining a type of the parallel Wi-Fi network or the parallel Wi-Fi access point, wherein the resolution is based on the type and the notification includes the type (step **306**). The type can be a same SSID, a different SSID, an evil twin, an evil brother, etc.

[0046] The steps can further include performing the detecting responsive to any of interference, initial turn up of the new Wi-Fi network, and any communication from the user including a phone call, a mobile application message to customer support, an email message, a response to a survey, and an in-person communication with an on-site technician (step **308**). That is, the process **300** can be implemented automatically, e.g., at the initial turn up of the new Wi-Fi network, as well as in response to problems, e.g., interference, the customer service call, etc.

[0047] The notification can include details for the user to turn off the parallel Wi-Fi network or the parallel Wi-Fi access point. The notification can include details for the user to remove profiles for the parallel Wi-Fi network or the parallel Wi-Fi access point in Wi-Fi client devices at the location. That is, the resolution flow of the process **300** is to remove the parallel network to prevent any issues. That is,

to notify a customer to turn off the parallel network, to notify the customer to remove profiles for parallel network from all devices at the location, to notify the carrier to disable the parallel network, to notify the carrier to contact customer to take action, and the like.

[0048] In an embodiment, the parallel Wi-Fi network or the parallel Wi-Fi access point has a same Service Set Identifier (SSID) as the new network. The detecting can be based on an access point in the new Wi-Fi network detecting another access point that is not part of the new Wi-Fi network but that has the same SSID. The detecting can be based on detecting Wi-Fi client devices disappearing and reappearing on the new Wi-Fi network, with at least some of the Wi-Fi client devices being non-mobile devices. Also, the detecting can be based on a combination of an access point in the new Wi-Fi network detecting another access point that is not part of the new Wi-Fi network but that has the same SSID, and detecting Wi-Fi client devices disappearing and reappearing on the new Wi-Fi network, with at least some of the Wi-Fi client devices being non-mobile devices.

[0049] In another embodiment, the parallel Wi-Fi network or the parallel Wi-Fi access point has a different Service Set Identifier (SSID) as the new network. The detecting can be based on an access point in the new Wi-Fi network detecting another access point that is not part of the new Wi-Fi network having a strong signal indicative of being at the same location. The detecting can be based on detecting Wi-Fi client devices disappearing and reappearing on the new Wi-Fi network, with at least some of the Wi-Fi client devices being non-mobile devices. Also, the detecting can be based on a combination of an access point in the new Wi-Fi network detecting another access point that is not part of the new Wi-Fi network having a strong signal indicative of being at the same location, and detecting Wi-Fi client devices disappearing and reappearing on the new Wi-Fi network, with at least some of the Wi-Fi client devices being non-mobile devices.

[0050] The process **300** can be performed by a controller communicatively coupled to the new Wi-Fi network via the Internet.

Parallel Network Resolution Flow

[0051] The objective of the process **300** is detection and resolution of any parallel network, i.e., the parallel Wi-Fi network or the parallel Wi-Fi access point. The resolution can include directly shutting off the parallel network, notifying the customer to turn off parallel network, notifying the customer to remove profiles for parallel network from devices in home, notifying the carrier to disable the parallel network, notifying the carrier to contact customer to take action, and the like. Also, there can be a post step of checking that the issue has been resolved at a later time. This can include performing the process **300** to detect if there is still a parallel network.

[0052] As mentioned above, the process **300** can be implemented remotely such as in the cloud **12**. The output of the process **300** can include notifications to the customer (i.e., the end user of the new Wi-Fi network) as well as the carrier (i.e., a service provider providing Internet access to the end user). Various different approaches for resolution are contemplated including actions by the end user, by the carrier, etc.

Alarms, Dashboards, and Support

[0053] Commonly-assigned U.S. patent application Ser. No. 17/071,015, filed Oct. 15, 2020, and entitled “Intelligent monitoring systems and methods for cloud-based Wi-Fi,” the contents of which are incorporated by reference in their entirety, describes various techniques for cloud-based Wi-Fi monitoring to alert, predict, and solve customer issues for service providers. The objective of this monitoring is to improve customer experience and proactively address issues with the goal of reducing customer churn. The parallel network described herein is one example of an issue that leads to poor Wi-Fi. The various techniques described herein are contemplated with the disclosure in U.S. patent application Ser. No. 17/071,015. That is, the parallel network can be a factor in the Call in Rate (CIR) predictor, customer churn, Net Promoter Score (NPS) predictor, etc.

[0054] The process 300 can also be utilized with alarms and a dashboard for visualization. For example, a Graphical User Interface (GUI) can provide a dashboard showing prevalence across networks of homes with suspected parallel networks (this can be part of the Monitor dashboard in U.S. patent application Ser. No. 17/071,015). The dashboard can include a breakdown by type, a listing of locations with issues (Alert), and support information. The listing of locations can be sorted or displayed by severity of type or severity of case as well as highlighting new networks that have just been discovered to have the issue.

[0055] The process 300 can also include support related features for the carrier including highlighting there is a suspected parallel network in the location in a Network Operations Center (NOC) display, prefilling a ticket with information about the parallel network, actually entering a ticket based on suspecting a parallel network, and the like.

[0056] It will be appreciated that some embodiments described herein may include one or more generic or specialized processors (“one or more processors”) such as microprocessors; Central Processing Units (CPUs); Digital Signal Processors (DSPs); customized processors such as Network Processors (NPs) or Network Processing Units (NPU), Graphics Processing Units (GPUs), or the like; Field Programmable Gate Arrays (FPGAs); and the like along with unique stored program instructions (including both software and firmware) for control thereof to implement, in conjunction with certain non-processor circuits, some, most, or all of the functions of the methods and/or systems described herein. Alternatively, some or all functions may be implemented by a state machine that has no stored program instructions, or in one or more Application-Specific Integrated Circuits (ASICs), in which each function or some combinations of certain of the functions are implemented as custom logic or circuitry. Of course, a combination of the aforementioned approaches may be used. For some of the embodiments described herein, a corresponding device in hardware and optionally with software, firmware, and a combination thereof can be referred to as “circuitry configured or adapted to,” “logic configured or adapted to,” etc. perform a set of operations, steps, methods, processes, algorithms, functions, techniques, etc. on digital and/or analog signals as described herein for the various embodiments.

[0057] Moreover, some embodiments may include a non-transitory computer-readable storage medium having computer readable code stored thereon for programming a computer, server, appliance, device, processor, circuit, etc.

each of which may include a processor to perform functions as described and claimed herein. Examples of such computer-readable storage mediums include, but are not limited to, a hard disk, an optical storage device, a magnetic storage device, a ROM (Read Only Memory), a PROM (Programmable Read Only Memory), an EPROM (Erasable Programmable Read Only Memory), an EEPROM (Electrically Erasable Programmable Read Only Memory), Flash memory, and the like. When stored in the non-transitory computer-readable medium, software can include instructions executable by a processor or device (e.g., any type of programmable circuitry or logic) that, in response to such execution, cause a processor or the device to perform a set of operations, steps, methods, processes, algorithms, functions, techniques, etc. as described herein for the various embodiments.

[0058] Although the present disclosure has been illustrated and described herein with reference to preferred embodiments and specific examples thereof, it will be readily apparent to those of ordinary skill in the art that other embodiments and examples may perform similar functions and/or achieve like results. All such equivalent embodiments and examples are within the spirit and scope of the present disclosure, are contemplated thereby, and are intended to be covered by the following claims. Moreover, it is noted that the various elements, operations, steps, methods, processes, algorithms, functions, techniques, etc. described herein can be used in any and all combinations with each other.

What is claimed is:

1. A method comprising:
 - identifying a Wi-Fi network at a location;
 - detecting, in association with the Wi-Fi network, an event associated with performance of the Wi-Fi network;
 - detecting a parallel Wi-Fi network at the location, the detection of the parallel Wi-Fi network comprising identification of a type of parallel Wi-Fi network; and
 - performing, based on the detection of the parallel Wi-Fi network, a turn up of the Wi-Fi network.
2. The method of claim 1, further comprising the event comprising at least one of interference or congestion related to the Wi-Fi network.
3. The method of claim 1, further comprising the event comprising an indication that a number of connected devices to the Wi-Fi network are below a threshold, the event indicating such devices being connected to the parallel Wi-Fi network.
4. The method of claim 1, further comprising the event comprising:
 - detecting client device connecting to the Wi-Fi network; and
 - detecting the client device disconnecting to the Wi-Fi network, the client device's disconnection corresponding to a connection with the parallel Wi-Fi network.
5. The method of claim 4, further comprising the detection of the event corresponding to the disconnection of the client device.
6. The method of claim 4, further comprising:
 - detecting the client device reconnecting to the Wi-Fi network, wherein the detection of the event is based on the reconnection.
7. The method of claim 1, further comprising:
 - the identification of the Wi-Fi network comprising identifying an access point for the Wi-Fi network; and

the detection of the parallel Wi-Fi network comprising identifying another access point providing the parallel Wi-Fi network.

8. The method of claim 7, further comprising the access point and the other access point comprising a same service set identifier (SSID).

9. The method of claim 7, further comprising the access point and the other access point comprising a different service set identifier (SSID).

10. The method of claim 1, further comprising: causing display of a dashboard, the dashboard comprising information indication locations associated with the Wi-Fi network and the parallel Wi-Fi network, the dashboard enabling a service provider to perform a resolution operation responsive to the event.

11. The method of claim 1, further comprising performing the detection of the event at an initial turn up of the Wi-Fi network.

12. The method of claim 1, further comprising the Wi-Fi network being provided by an access point supporting at least two wireless channels, such that client links and backhaul links do not share a same channel.

13. The method of claim 1, further comprising: determining, based at least on the detection of the parallel Wi-Fi network, a resolution related to the Wi-Fi network and the parallel Wi-Fi network; and performing the resolution, the resolution removing access to the parallel Wi-Fi network at the location, the resolution comprising at least one of disabling the parallel Wi-Fi network or removing a profile of the parallel Wi-Fi network.

14. The method of claim 1, further comprising communicating a notification to at least one of a user or service provider indicating at least the event.

15. A system comprising: a processor; and memory comprising instructions that, when executed, cause the processor to:

identify a Wi-Fi network at a location;

detect, in association with the Wi-Fi network, an event associated with performance of the Wi-Fi network; detect a parallel Wi-Fi network at the location, the detection of the parallel Wi-Fi network comprising identification of a type of parallel Wi-Fi network; and

perform, based on the detection of the parallel Wi-Fi network, a turn up of the Wi-Fi network.

16. The system of claim 15, wherein the event comprise at least one of interference or congestion related to the Wi-Fi network.

17. The system of claim 15, wherein the processor is further caused to:

detect client device connecting to the Wi-Fi network; and detect the client device disconnecting to the Wi-Fi network, the client device's disconnection corresponding to a connection with the parallel Wi-Fi network.

18. The system of claim 17, wherein the detection of the event corresponds to the disconnection of the client device.

19. The system of claim 17, wherein the processor is further caused to:

detect the client device reconnecting to the WiFi network, wherein the detection of the event is based on the reconnection.

20. A non-transitory computer-readable storage medium tangibly encoded with computer-executable instructions, that when executed by a processor, perform a method comprising:

identifying a Wi-Fi network at a location; detecting, in association with the Wi-Fi network, an event associated with performance of the Wi-Fi network; detecting a parallel Wi-Fi network at the location, the detection of the parallel Wi-Fi network comprising identification of a type of parallel Wi-Fi network; and performing, based on the detection of the parallel Wi-Fi network, a turn up of the Wi-Fi network.

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